

Sept 5, 2025

Synthesis for ICs (integrated circuits)

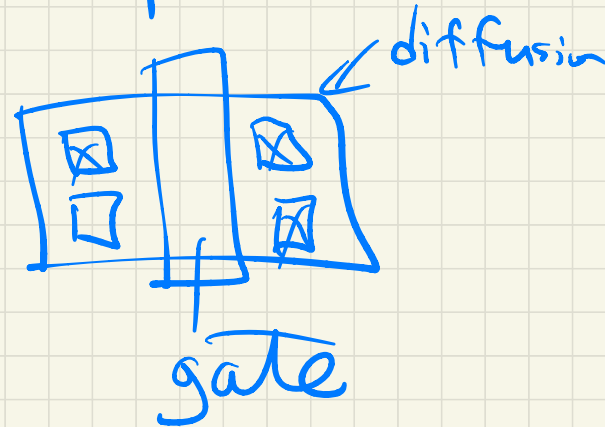
IC media: different kinds of ICs
could be the end target of
CAD flow

① Full-custom VLSI

- drawing rectangles/routers by hand

- rarely done these days. exceptions:

- FPGA logic/routing arch
- analog / RF - RAM cells



② Semi-custom VLSI

- standard cells
- library of cells provided by foundary (eg TSMC, ST Micros...)
-  Inverter, NAND, XOR,
- multiple versions of each cells with different sizes (different drive strengths)
- Each cell has a layout

Output: all mask layers

III Field-programmable

- FPGAs
- CGRAs (coarse-grained reconfigurable arrays)

Output: Configuration bitstream

We will stray between all
three.

Optimization is key part of CAD

What is optimization?

- $\exists$ many ways to implement same function
 - Some have good area
 - " " " delay
 - " " " power
 - Typically trade-offs between these
- Optimization ~ is a way to search thru some (all?) choices to find one you want → target speed, good area

Optimization Strategies

- simulated annealing
- graph search
- genetic algorithms
- linear programming
- convex optimizations
- reinforcement learning
- gradient descent
- greedy algorithm
- branch-and-bound
- ant colony optimization

- dynamic programming
- network flow

CAD Sub-tasks:

- many sub-tasks have evolved in the synthesis of ICs
- we will cover main subtasks in ECE1387

Top-to-Bottom View of Synthesis Tasks

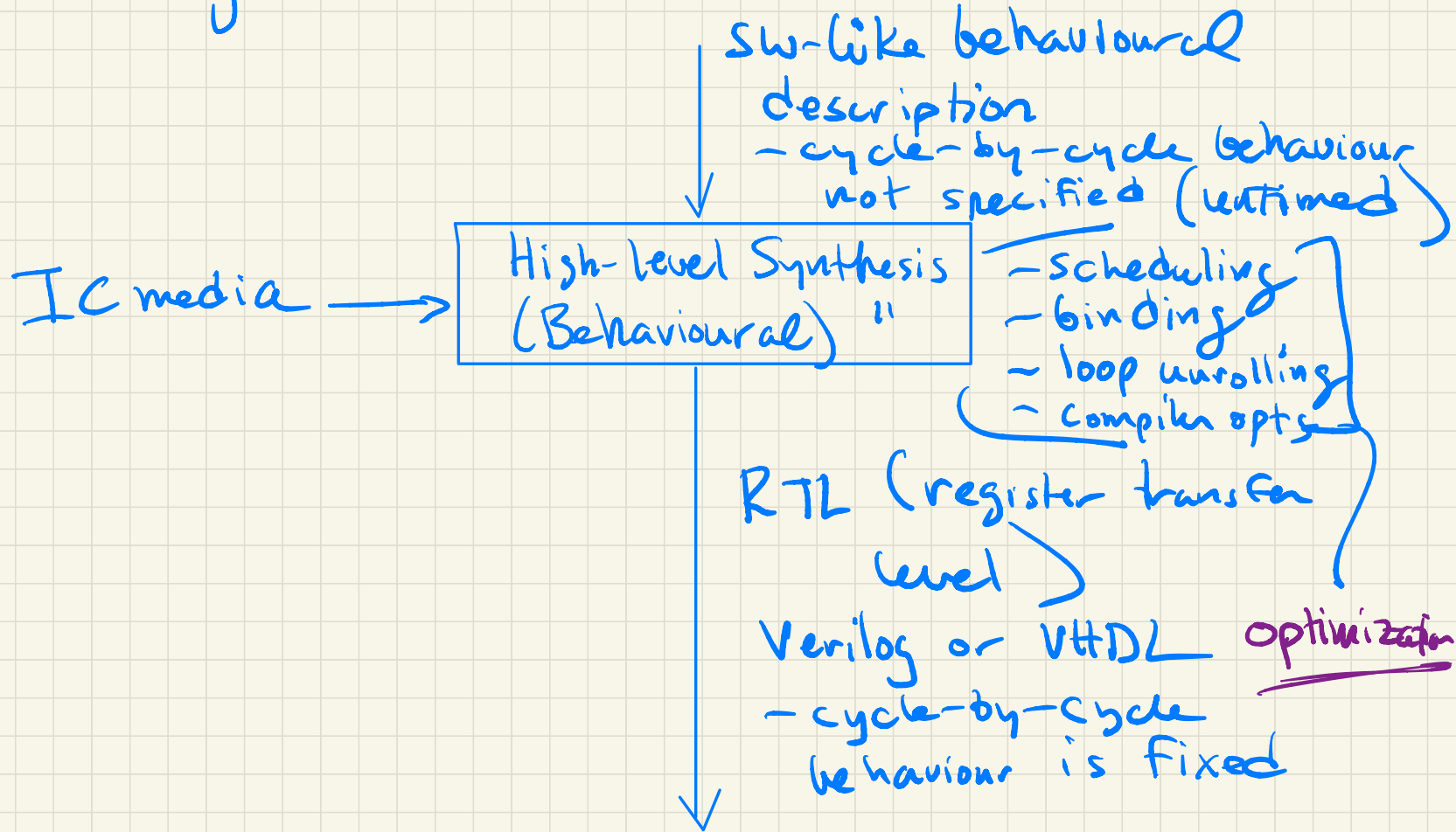
Input: Desired behaviour of chip @
behavioural (Software-like) level

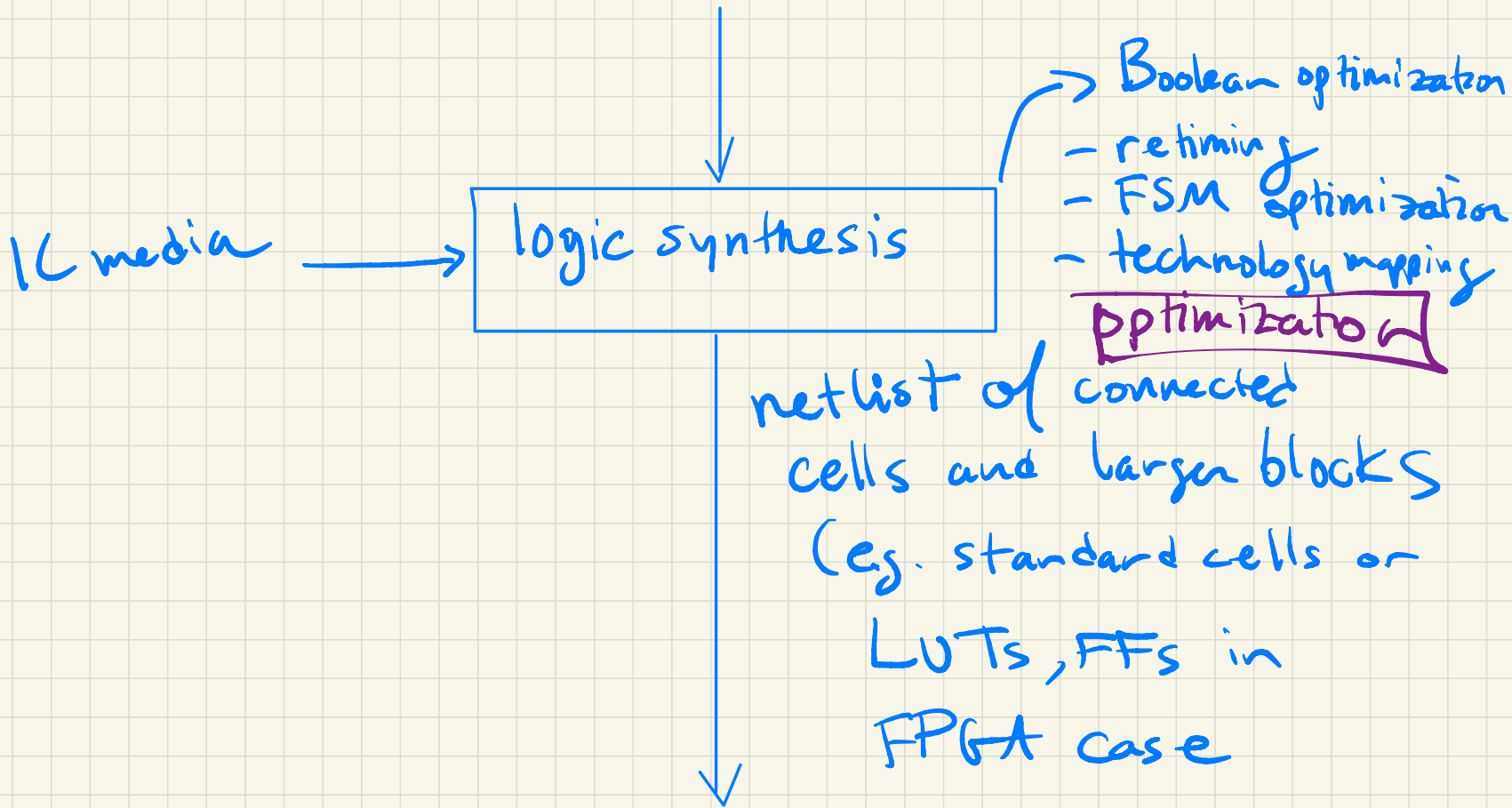
- speed spec
 - area spec
 - power spec
- Constraints

Output:  

- all mask layers (std cells)
- config bitstream (field prog.)

3 major steps:





IC media



layout synthesis



- partitioning
 - floorplanning
 - placement
 - routing
- optimization

all mask layers or
config bitstream



Course Strategy

IC
media

CAD
task

optimization
strategy

pick one of
each and
describe in
depth -